

Benchmarking & evaluating systems

CS524: Big Data Visualization & Analytics
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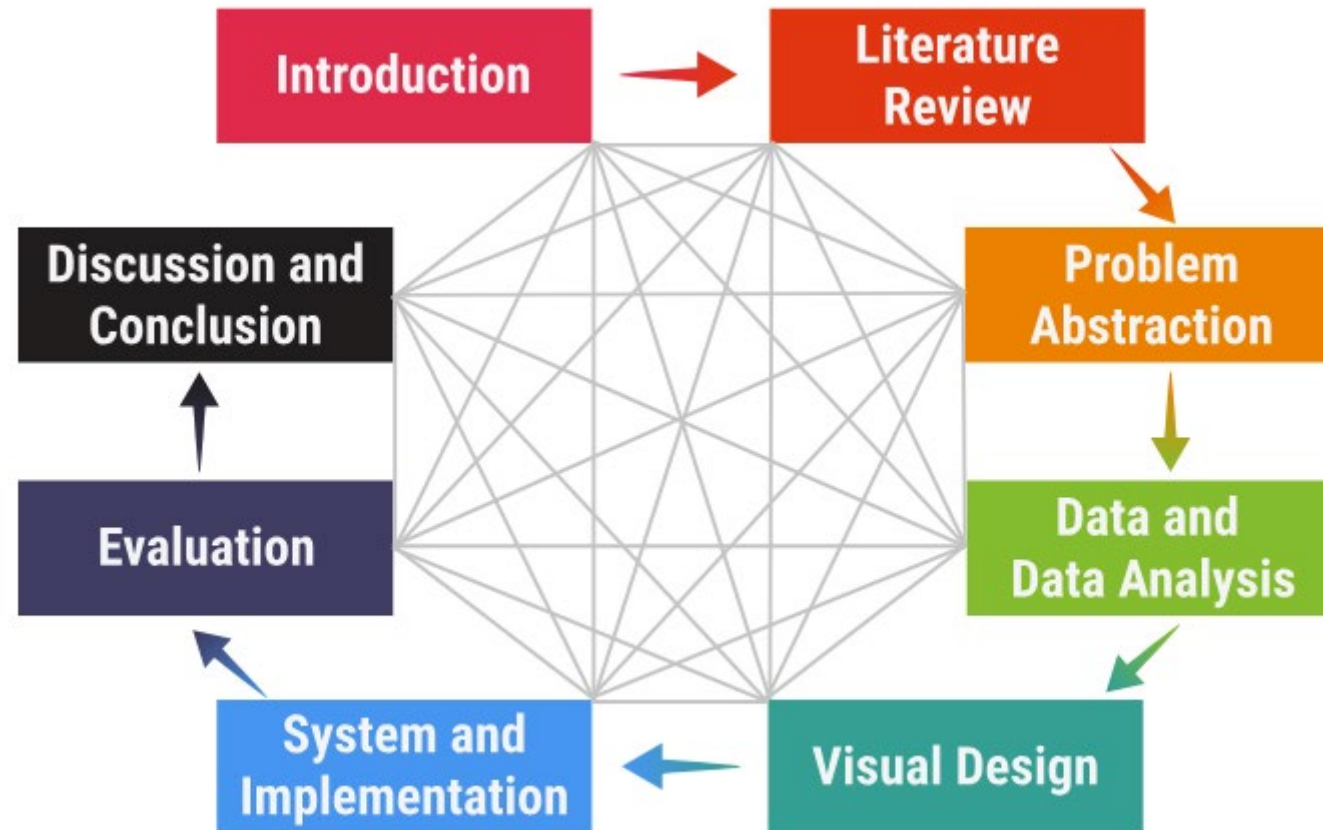
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First, the components of a VA system manuscript



Wu et al., 2023

Evaluation of VA systems

Chen and Ebert, 2019

Khayat et al., 2020

Wu et al., 2023

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Barographics Conference on Visualization (BaroVis) 2019
M. Glöckner, H. Leito, and L. Viola
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An Ontological Framework for Supporting the Design and Evaluation of Visual Analytics Systems

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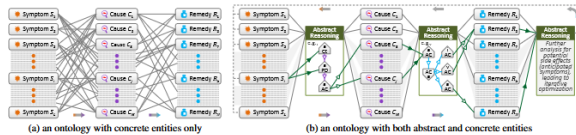


Figure 1: An ontology can support the design and evaluation of visual analytics systems by enabling systematic recording of symptoms, causes, remedies, and side-effects, facilitating the analysis of causal relations, and comparison of different design options. (a) An ontology that records only concrete entities (e.g., instances in practice) will take years and decades to build. (b) An ontology built around abstract entities can enable abstract reasoning in a scalable manner while allowing gradual addition of concrete entities.

Abstract

Designing, evaluating, and improving visual analytics (VA) systems is a primary area of activities in our discipline. In this paper, we present an ontological framework for recording and categorizing technical shortcomings to be addressed in a VA workflow, reasoning about the causes of such problems, identifying technical solutions, and anticipating secondary effects of the solutions. The methodology is built on the theoretical premise that designing a VA workflow is an optimization of the cost-benefit ratio of the processes in the workflow. It makes uses three fundamental measures to group and connect "symptoms", "causes", "remedies", and "side-effects", and guide the search for potential solutions to the problems. In terms of requirement analysis and system design, the proposed methodology can enable system designers to explore the decision space in a structured manner. In terms of evaluation, the proposed methodology is time-efficient and complementary to various forms of empirical studies, such as user surveys, controlled experiments, observational studies, focus group discussions, and so on. In general, it reduces the amount of trial-and-error in the lifecycle of VA system development.

1. Introduction

Designing, evaluating, and improving visual analytics (VA) systems is a primary area of activities in our discipline. Since the first IEEE VAST in 2006, a large number of system papers and application papers have been published, signifying the important contributions to the technology advancements as well as a variety of successful applications [WHC⁺17]. During this relatively short period, VA researchers have made noticeable progress in adopting some best practices in software engineering (e.g., [Sum15]). Today, most system papers and application papers in VA conferences reported user-centered requirement analysis and evaluation. Some

reported the use of incremental, iterative, nested, agile, and evolutionary development approaches [Lar03, Mun09, Mar14].

While these approaches have no doubt successfully delivered many VA systems and applications, assessing and sharing the experience of developing these VA systems and applications can be difficult. These difficulties include:

- The step made from requirements analysis to design specification is typically based on intuitions and the design decisions made are often undocumented, difficult to share, or not reproducible in another application.
- The evaluation based on individual use cases is sometimes per-

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The Validity, Generalizability and Feasibility of Summative Evaluation Methods in Visual Analytics

Mosab Khayat, Morteza Karimzadeh, David S. Ebert, Fellow, IEEE, Arif Ghafor, Fellow, IEEE

Abstract—Many evaluation methods have been used to assess the usefulness of Visual Analytics (VA) solutions. These methods stem from a variety of origins with different assumptions and goals, which cause confusion about their proofing capabilities. Moreover, the lack of discussion about the evaluation processes may limit our potential to develop new evaluation methods specialized for VA. In this paper, we present an analysis of evaluation methods that have been used to summatively evaluate VA solutions. We provide a survey and taxonomy of the evaluation methods that have appeared in the VAST literature in the past two years. We then analyze these methods in terms of validity and generalizability of their findings, as well as the feasibility of using them. We propose a new metric called summative quality to compare evaluation methods according to their ability to prove usefulness, and make recommendations for selecting evaluation methods based on their summative quality in the VA domain.

Index Terms—Summative evaluation, usefulness, evaluation process, taxonomy, visual analytics

1 INTRODUCTION

Visual analytics (VA) solutions emerged in the past decade and tackled many problems in a variety of domains. The power of combining the abilities of human and machine creates fertile ground for new solutions to grow. However, the rise of these hybrid solutions complicates the process of evaluation. Unlike automated algorithms, the behavior of visual analytics solutions depends on the user who operates them. This creates a new dimension of variability in the performance of the solutions that needs to be accounted for in evaluation. The existence of a human in the loop, however, allows researchers to borrow evaluation methods from other domains, such as sociology [33], to extract information with the help of the user. Such evaluation methods allow developers to assess their solutions even when a formal summative evaluation is not feasible. The challenge in these methods, however, lies in gathering and analyzing qualitative data to build valid evidence. Many methods have been used to evaluate VA solutions, each originally developed to answer different questions with different evaluation intentions, including formative, summative and exploratory [2]. Nevertheless, many of these methods have been extensively applied in summative evaluation despite the fact that some are only suitable for formative or exploratory assessment, not summative evaluation.

In this paper, we survey and analyze the evaluation methods commonly used with summative intentions in VA research. Specifically, we survey the papers presented at VAST 2017 and 2018, resulting in a seven-category taxonomy of evaluation methods. We identify the activities typically performed within each category, focusing on the activities that could introduce risks to the validity and the generalizability of the methods' findings, and we use both of these factors to define summative quality. We also define feasibility based on the identified activities and the limitations in applying evaluation methods in various scenarios. Finally, we use summative quality and feasibility to compare summative evaluation methods. Unlike existing problem-driven prescriptions [53], we analyze the risks to the validity, generalizability, and feasibility of each evaluation method by focusing on the activities employed in each method. We then provide a prescription of evaluation methods based on (a) their ability to prove usefulness and (b) their feasibility.

The contributions of this paper can be summarized as follows:

- A survey, taxonomy and risk-based breakdown and analysis of evaluation methods used in summative evaluation of VA solutions.
- A summative quality metric to assess the summative quality of evaluation methods based on the potential risks to the validity and the generalizability of the methods' findings.
- An analysis and prescription of summative evaluation methods in terms of their summative quality and feasibility.

This paper is organized as follows: In Section 2, we provide important definitions, and review related work in Section 3. In Section 4, we present our survey of evaluation methods used for summative assessment. We analyze these methods in Section 5, followed by a set of recommendations for practitioners in Section 6. Finally, we conclude the paper and provide directions for future research.

2. USEFULNESS AND SUMMATIVE EVALUATION DEFINITIONS

The term "summative assessment" has roots in the field of education, which distinguishes it from formative assessment [77]. The former assesses students objectively at the end of a study period using standardized exams, while the latter focuses on the learning process and the students' progress in meeting standards. In visualization and the literature, summative evaluation has been traditionally referred to as a type of studies that measures the quality of a developed system using methods such as formal lab experiments [53]. This is in contrast with formative assessment, which seeks to inform the design and development processes by applying techniques such as expert feedback [56].

There have been some suggestions in the literature that evaluation in tension should be unlinked from evaluation methods. Ellis and Dix [26] argue that a formal lab experiment of a completely developed system can be conducted with formative intentions to suggest improvement. On the other hand, Munzner [53] argues that formative methods such as expert feedback can be used with summative intentions to validate the outcome of design decisions. We agree with these arguments and believe that it is essential to give a formal definition of summative evaluation as an intention rather than an evaluation stage.

From the discussions in [77], we define summative evaluation as a systematic process which generates evidence about the degree of accomplishment of the given objectives (standards) for an assessed object (a solution) at a point in time. We use the term "solution" throughout this article to refer to different approaches for tackling a problem. VA research covers different types of solutions ranging from algorithms, to visualizations, to the integration of these in a holistic system (See [68] for design study contributions). "Standards" refer to benchmarks that are used to distinguish useful solutions from non-useful ones. These benchmarks are determined during the requirement elicitation stage, e.g. by conducting qualitative interviews with domain experts.

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In Defence of Visual Analytics Systems: Replies to Critics

Aoyu Wu, Dazhen Deng, Furui Cheng, Yingcai Wu, Shixia Liu, and Huamin Wu

Abstract—The last decade has witnessed many visual analytics (VA) systems that make successful applications to wide-ranging domains like urban analytics and explainable AI. However, their research rigor and contributions have been extensively questioned within the visualization community. We come in defence of VA systems by contributing two interview studies for gathering critics and responses to those criticisms. First, we interview 24 researchers to collect criticisms the review comments on their VA work. Through an iterative coding and refinement process, the interview feedback is summarized into a list of 36 common criticisms. Second, we interview 17 researchers to validate our list and collect their responses, thereby discussing implications for defending and improving the scientific values and rigor of VA systems. We highlight that the presented knowledge is deep, extensive, but also imperfect, provocative, and controversial, and thus recommended reading with an inclusive and critical eye. We hope our work can provide thoughts and foundations for conducting VA research and spark discussions to promote the research field forward more rigorously and vibrantly.

Index Terms—Visual Analytics, Theory, Qualitative Study, Design Study, Application, Theoretical and Empirical Research

1 INTRODUCTION

The last decades have witnessed extensive and ever-growing research interests on visual analytics. By combining automated data analysis techniques with interactive visualizations, visual analytics facilitates an effective understanding, reasoning, and decision-making on large and complex datasets [24]. Since the IEEE Conference on Visual Analytics Science and Technology (VAST) was founded in 2006, researchers have contributed many systems for solving complex problems in wide-ranging applications, to which we refer as VA systems. Many systems are outcomes of problem-driven research, whose values have been demonstrated through successful applications in high-impact domains, such as social media [72], sports [2, 73], urban analytics [16, 35], and explainable AI [37, 49]. Research on VA systems have also become an important and impactful research field in visualization (Fig. 1).

Despite the success of those application-oriented VA systems, their research contributions and rigor have been extensively challenged, discussed, and debated [7, 41, 55, 70]. Underlying those debates is the tension between the impetus to create specific software artifacts and the drive of academic research to produce general knowledge [47]. This tension raises the frequently asked question of "what our visualization community can learn from the VA system beyond solving the domain-specific problem". Furthermore, the design and validation methodology of VA systems is human centered and thus qualitative and subjective in nature. This nature might contrast with the tradition in science and computer science that embraces quantification and objectivity [23, 41].

The above discrepancies point out some practical problems in the research field. For contributors, their process of planning, developing, validating, and reporting VA systems is prone to diverse mistakes or pitfalls at different stages [55]. The process is challenging because conducting research on VA systems requires a wide range of skills, such as leveraging HCI approaches to understand target users, implementing automated algorithms, and designing effective visual design [62]. For reviewers, assessing the quality of VA system research requires judging and weighing the above aspects. This assessing process tends to be subjective due to the lack of a shared ground among reviewers regarding

objective criteria for evaluating VA systems [70].

We, therefore, seek to identify a common space of critics for VA systems and corresponding replies to critics. We note that existing instructions on assessing the broad scope of visualization research are not specific to VA systems [33]. Besides, the current reflections and discussions on the rigor of visualization design techniques (e.g., [40, 41, 55]) are mainly constricted from authors' engagement and experiences. While those discussions are undoubtedly critical and insightful, we argue that surveying a broader scope of VA researchers will lead to a more diverse, representative, and convincing understanding.

We conduct two interview studies to reflect on VA systems in terms of their common criticisms and corresponding replies. As shown in Fig. 2, we start by interviewing 24 researchers in charge of 47 VA systems by asking them, "what are the criticisms you have received during peer-reviewing?" We obtain 257 instances of criticisms, which are further classified into 36 common types through iterative categorization. As those criticisms are diverse, deep, and challenging, we conduct another interview study with 17 researchers to validate our classification and gather their replies, e.g., how to mitigate and respond to those criticisms? Drawing upon replies to low-level criticisms, we discuss implications for a high-level question - how to conduct research to defend and improve the research values and rigor of VA systems?

We position our work as a preliminary probe into assessment criteria for VA systems. We expect our work to benefit both VA contributors and reviewers by offering an evidence-based reference for assessing VA systems. We, however, emphasize that criticisms are intrinsically biased, changing, controversial, and fallible, so is our result. It is not our intention to advocate a golden template, but to construct a preliminary and debatable set of criteria for assessing VA systems. We hope our work can provide foundations and spark discussions to make the research field more rigorous and vibrant. We make our interview data and other supplemental material available at re-vast.github.io.

2 RELATED WORK

We focus on research that primarily contributes to VA system for an application. Thus, we review existing literature regarding theoretical advances in visualization application and design study, visual analytics, and empirical methods for understanding the field of visualization.

2.1 Visualization Application and Design Study

Sudaimir et al. [55] formally introduced a visualization design study as "a project in which visualization researchers analyze a specific real-world problem faced by domain experts, design a visualization system that supports solving this problem, validate the design, and reflect about lessons learned in order to refine visualization design guidelines". Following this definition, they further proposed a nine-stage methodology framework for visualization design studies. Their well-cited framework has become a common method of developing

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DEPARTMENT: VISUALIZATION VIEWPOINTS

Grand Challenges in Visual Analytics Applications

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In the past two decades, research in visual analytics (VA) applications has made tremendous progress, not just in terms of scientific contributions, but also in real-world impact across wide-ranging domains including bioinformatics, urban analytics, and explainable AI. Despite these success stories, questions on the rigor and value of VA application research have emerged as a grand challenge. This article outlines a research and development agenda for making VA application research more rigorous and impactful. We first analyze the characteristics of VA application research and explain how they cause the rigor and value problem. Next, we propose a research ecosystem for improving scientific value, and rigor and outline an agenda with 12 open challenges spanning four areas, including foundation, methodology, application, and community. We encourage discussions, debates, and innovative efforts toward more rigorous and impactful VA research.

Visual analytics (VA) applications leverage software artifacts with VA techniques to solve real-world problems. They combine automated data analysis techniques with interactive visualizations to facilitate effective reasoning and decision-making on large and complex datasets. Since the establishment of the IEEE Conference on Visual Analytics Science and Technology in 2006, VA applications have grown into a significant and influential research field in visualization. For example, research on VA applications

occupies about 34% (41/120) of the full papers published in the IEEE Visualization Conference (IEEE VIS) 2022. VA applications have also achieved substantial societal impact by providing success stories on solving real-world problems, especially in wide-ranging domains, such as bioinformatics, urban analytics, explainable artificial intelligence (AI), and social media. Nevertheless, decades of research and practices of VA applications have exposed multiple value- and rigor-related questions (see Figure 1). In light of these questions, we organized a panel at IEEE VIS 2022 to discuss grand challenges for making this research field more valuable and rigorous. The panel theme was informed by an informal interview study with 17 active researchers in VA system research.²

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Mosab Khayat, Morteza Karimzadeh, David S. Ebert, Fellow, IEEE, Arif Ghafoor, Fellow, IEEE

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Index Terms—Summative evaluation, usefulness, evaluation process, taxonomy, visual analytics

1 INTRODUCTION

Visual analytics (VA) solutions emerged in the past decade and tackled many problems in a variety of domains. The power of combining the abilities of human and machine creates fertile ground for new solutions to grow. However, the rise of these hybrid solutions complicates the task of evaluating them.

- A survey, taxonomy and risk-based breakdown and analysis of evaluation methods used in summative evaluation of VA solutions.
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“The challenge in these methods, however, lies in gathering and analyzing qualitative data to build valid evidence.”

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Index Terms—Visual Analytics, Theory, Qualitative Study, Design Study, Application, Theoretical and Empirical Research

“Evaluating visualization systems has been traditionally difficult, and it is arguably even harder to evaluate complex VA systems. To guide our discussions, we start with surveying existing valuation methods for VA systems (see the supplemental material for details).”

automated algorithms, and designing effective visual designs [62]. For reviewers, assessing the quality of VA system research requires judging and weighing the above aspects. This assessing process tends to be subjective due to the lack of a shared ground among reviewers regarding

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DEPARTMENT: VISUALIZATION VIEWPOINTS

Grand Challenges in Visual Analytics Applications

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“... the evaluation of VA applications is sometimes perceived as anecdotal evidence, making it difficult for readers to assess their research value.”

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Grand Challenges in Visual Analytics Applications

How can I replicate this system?
Are they open-sourced?

How to fit the system into my
data analysis workflow?

What general knowledge can
the visualization community
learn from ad-hoc systems?



The visualizations look fancy but
complex. I stared at them but failed
to figure out what they could tell me.

The evaluation method is informal.
How to quantify the measurement?

What is the research theory and
methodology in this field?

on large and complex datasets.¹ Since the establishment of the IEEE Conference on Visual Analytics Science and Technology in 2006, VA applications have grown into a significant and influential research field in visualization. For example, research on VA applications

impact domains, such as bioinformatics, urban analytics, explainable artificial intelligence (AI), and social media.

Nevertheless, decades of research and practices of VA applications have exposed multiple value- and rigor-related questions (see Figure 1). In light of these questions, we organized a panel at IEEE VIS 2022 to discuss grand challenges for making this research field more valuable and rigorous. The panel theme was informed by an informal interview study with 17 active researchers in VA system research.²

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Evaluation of VA systems

Wu et al., 2023

“... the evaluation of VA applications is sometimes perceived as anecdotal evidence, making it difficult for readers to assess their research value.”

- A survey of evaluation methods of VA solutions suggests that 44% of studies performed **summative evaluation** without any human subjects (e.g., via application scenarios).
- VA solutions ask for new evaluation strategies because the classic quantitative and qualitative or HCI approaches are not applicable; we need to evaluate the visual information discovery process, which is heavily 1) user, data, and task-specific as well as 2) process-oriented.

EDITOR: Theresa-Marie Rhyne, theresamarietyne@gmail.com

DEPARTMENT: VISUALIZATION VIEWPOINTS

Grand Challenges in Visual Analytics Applications

Aoyu Wu , Harvard University, Cambridge, MA, 02138, USA

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In the past two decades, research in visual analytics (VA) applications has made tremendous progress, not just in terms of scientific contributions, but also in real-world impact across wide-ranging domains including bioinformatics, urban analytics, and explainable AI. Despite these success stories, questions on the rigor and value of VA application research have emerged as a grand challenge. This article outlines a research and development agenda for making VA application research more rigorous and impactful. We first analyze the characteristics of VA application research and explain how they cause the rigor and value problem. Next, we propose a research ecosystem for improving scientific value, and rigor and outline an agenda with 12 open challenges spanning four areas, including foundation, methodology, application, and community. We encourage discussions, debates, and innovative efforts toward more rigorous and impactful VA research.

Visual analytics (VA) applications leverage software artifacts with VA techniques to solve real-world problems. They combine automated data analysis techniques with interactive visualizations to facilitate effective reasoning and decision-making on large and complex datasets.¹ Since the establishment of the IEEE Conference on Visual Analytics Science and Technology in 2006, VA applications have grown into a significant and influential research field in visualization. For example, research on VA applications

occupies about 34% (41/120) of the full papers published in the IEEE Visualization Conference (IEEE VIS) 2022. VA applications have also achieved substantial societal impact by providing success stories on solving real-world problems, especially in wide-ranging high-impact domains, such as bioinformatics, urban analytics, explainable artificial intelligence (AI), and social media.

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- **Summative evaluations:** *after* design, to judge the tool (e.g., prove usefulness, quantify benefits, compare).
- **Formative evaluations:** *during* the design, to improve the tool (e.g., find usability problems, refine interactions).

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“... the evaluation of VA applications is sometimes perceived as anecdotal evidence, making it difficult for readers to assess their research value.”

- Why users prefer a complex visual interface enhanced by an active learning-based approach instead of combining a much simpler user interface with an automatic approach?
- What are the additional benefits of the carefully designed interactive visualization? Or, would users be willing to spend a lot of time interacting with the complex interface?

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Summative evaluation methods

Khayat et al., 2020

- A survey, taxonomy and risk-based breakdown and analysis of evaluation methods used in summative evaluation of VA solutions.

“One argument made by Munzner was the necessity of summative evaluation during each stage of design studies to evaluate the outcome of that individual stage. Sedlmair et al. and Mckenna et al. made similar arguments while describing the process of design studies.”

IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 26, NO. 1, JANUARY 2020

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The Validity, Generalizability and Feasibility of Summative Evaluation Methods in Visual Analytics

Mosab Khayat, Morteza Karimzadeh, David S. Ebert, *Fellow, IEEE*, Arif Ghafoor, *Fellow, IEEE*

Abstract—Many evaluation methods have been used to assess the usefulness of Visual Analytics (VA) solutions. These methods stem from a variety of origins with different assumptions and goals, which cause confusion about their proofing capabilities. Moreover, the lack of discussion about the evaluation processes may limit our potential to develop new evaluation methods specialized for VA. In this paper, we present an analysis of evaluation methods that have been used to summatively evaluate VA solutions. We provide a survey and taxonomy of the evaluation methods that have appeared in the VAST literature in the past two years. We then analyze these methods in terms of validity and generalizability of their findings, as well as the feasibility of using them. We propose a new metric called summative quality to compare evaluation methods according to their ability to prove usefulness, and make recommendations for selecting evaluation methods based on their summative quality in the VA domain.

Index Terms—Summative evaluation, usefulness, evaluation process, taxonomy, visual analytics

1 INTRODUCTION

Visual analytics (VA) solutions emerged in the past decade and tackled many problems in a variety of domains. The power of combining the abilities of human and machine creates fertile ground for new solutions to grow. However, the rise of these hybrid solutions complicates the process of evaluation. Unlike automated algorithmic solutions, the behavior of visual analytics solutions depends on the user who operates them. This creates a new dimension of variability in the performance of the solutions that needs to be accounted for in evaluation. The existence of a human in the loop; however, allows researchers to borrow evaluation methods from other domains, such as sociology [33], to extract information with the help of the user. Such evaluation methods allow developers to assess their solutions even when a formal summative evaluation is not feasible. The challenge in these methods, however, lies in gathering and analyzing qualitative data to build valid evidence.

Many methods have been used to evaluate VA solutions, each originally developed to answer different questions with different evaluation intentions, including formative, summative and exploratory [2]. Nevertheless, many of these methods have been extensively applied in summative evaluation despite the fact that some are only suitable for formative or exploratory assessment, not summative evaluation.

In this paper, we survey and analyze the evaluation methods commonly used with summative intentions in VA research. Specifically, we survey the papers presented at VAST 2017 and 2018, resulting in a seven-category taxonomy of evaluation methods. We identify the activities typically performed within each category, focusing on the activities that could introduce risks to the validity and the generalizability of the methods' findings, and we use both of these factors to define summative quality. We also define feasibility based on the identified activities and the limitations in applying evaluation methods in various scenarios. Finally, we use summative quality and feasibility to compare summative evaluation methods. Unlike existing problem-driven prescriptions [53], we analyze the risks to the validity, generalizability, and feasibility of each evaluation method by focusing on the activities employed in each method. We then provide a prescription of evaluation methods based on (a) their ability to prove usefulness and (b) their feasibility.

The contributions of this paper can be summarized as follows:

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- Morteza Karimzadeh is with the University of Colorado Boulder (formerly at Purdue University). Email: karimzadeh@colorado.edu.

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- A survey, taxonomy and risk-based breakdown and analysis of evaluation methods used in summative evaluation of VA solutions.

- A summative quality metric to assess the summative quality of evaluation methods based on the potential risks to the validity and the generalizability of the methods' findings.

- An analysis and prescription of summative evaluation methods in terms of their summative quality and feasibility.

This paper is organized as follows: In Section 2, we provide important definitions, and review related work in Section 3. In Section 4, we present our survey of evaluation methods used for summative assessment. We analyze these methods in Section 5, followed by a set of recommendations for practitioners in Section 6. Finally, we conclude the paper and provide directions for future research.

2 USEFULNESS AND SUMMATIVE EVALUATION DEFINITIONS

The term “summative assessment” has roots in the field of education, which distinguishes it from formative assessment [77]. The former assesses students objectively at the end of a study period using standardized exams, while the latter focuses on the learning process and the students' progress in meeting standards. In visualization and VA literature, summative evaluation has been traditionally referred to as the type of studies that measures the quality of a developed system using methods such as formal lab experiments [53]. This is in contrast with formative assessment, which seeks to inform the design and development processes by applying techniques such as expert feedback [56].

There have been some suggestions in the literature that evaluation intention should be unlinked from evaluation methods. Ellis and Dix [26] argue that a formal lab experiment of a completely developed system can be conducted with formative intentions to suggest improvement. On the other hand, Munzner [53] argues that formative methods such as expert feedback can be used with summative intentions to validate the outcome of different design stages. We agree with these arguments and believe that it is essential to give a formal definition of summative evaluation as an intention rather than an evaluation stage.

From the discussions in [77], we define summative evaluation as a systematic process which generates evidence about the degree of accomplishment of the given objectives (standards) for an assessed object (a solution) at a point in time. We use the term “solution” throughout this article to refer to different approaches for tackling a problem. VA research covers different types of solutions ranging from algorithms, to visualizations, to the integration of these in a holistic system (See [68] for design study contributions). “Standards” refer to benchmarks that are used to distinguish useful solutions from non-useful ones. These are commonly determined during the requirement elicitation stage, e.g., by conducting qualitative inquiries with domain experts.

Summative evaluation methods

Theoretical methods

- Logically derive evidence.
- E.g., algorithmic complexity analysis

Quantitative user testing

- Quantitatively estimate the performance of human subjects

Quantitative user opinion

- Assess subjective aspects
- E.g., structured questionnaires

Quantitative automation testing

- Objectively assess human-independent solutions.
- E.g., cross-validation

Insight-based evaluations

- Human subjects qualitatively identify a set of insights that can be reached with the help of a solution.

Case studies

- Qualitative methods which allow researchers to determine objective values and subjective opinions about the evaluated solution by interacting with human subjects who are typically domain experts.

Inspection methods

- Assess objective or subjective potentials of a solution without testing or recruiting human subject.
- E.g., requirement inspection.

Summative evaluation methods

- **Validity:**

Are the study findings correct?

- **Generalizability**

Do the findings apply beyond the specific cases studied?

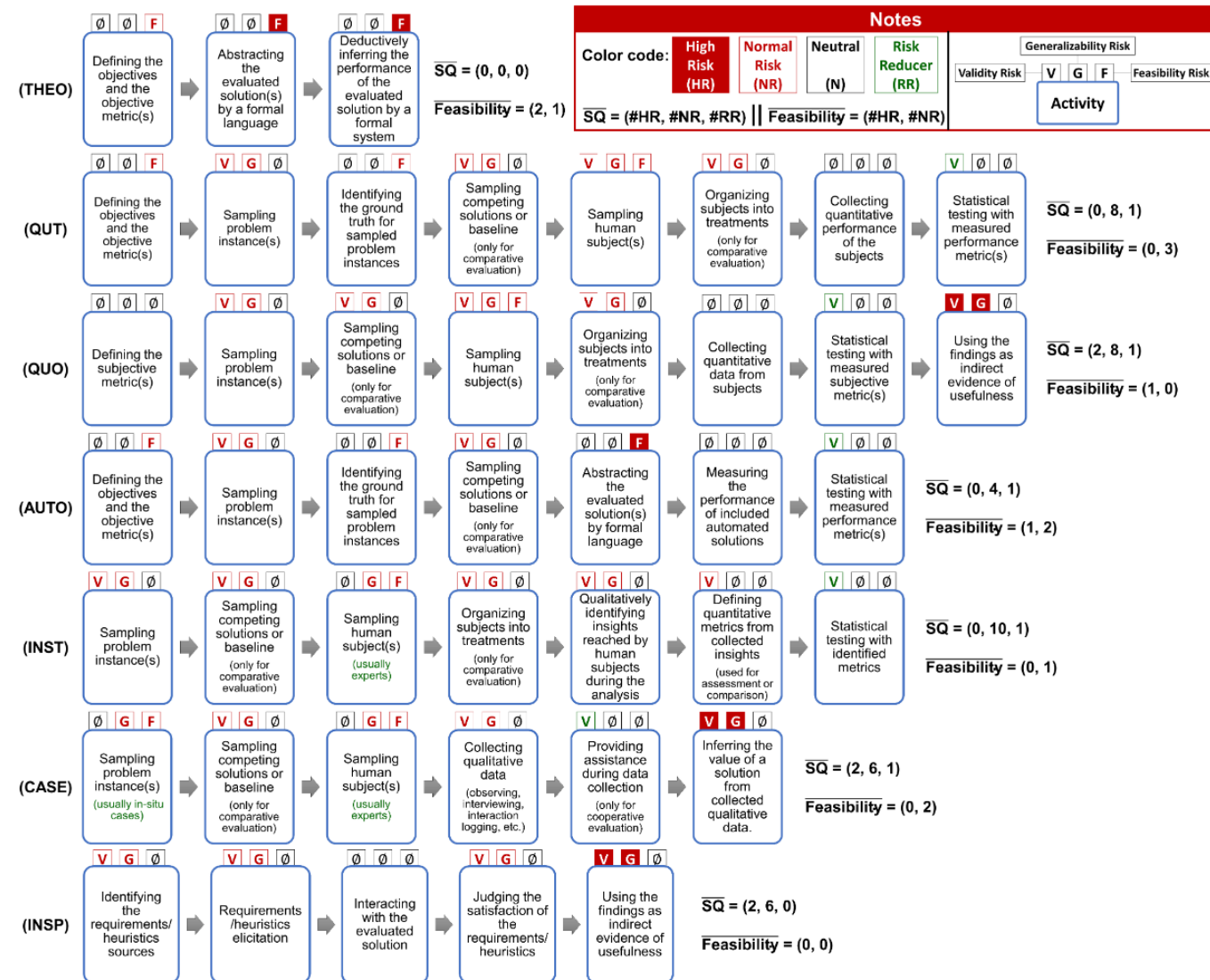
- **Feasibility:**

Can you actually do it?

Validity and generalizability into a single notion of summative quality: how likely is this method to produce evidence that's both correct and broadly applicable?



Summative evaluation methods



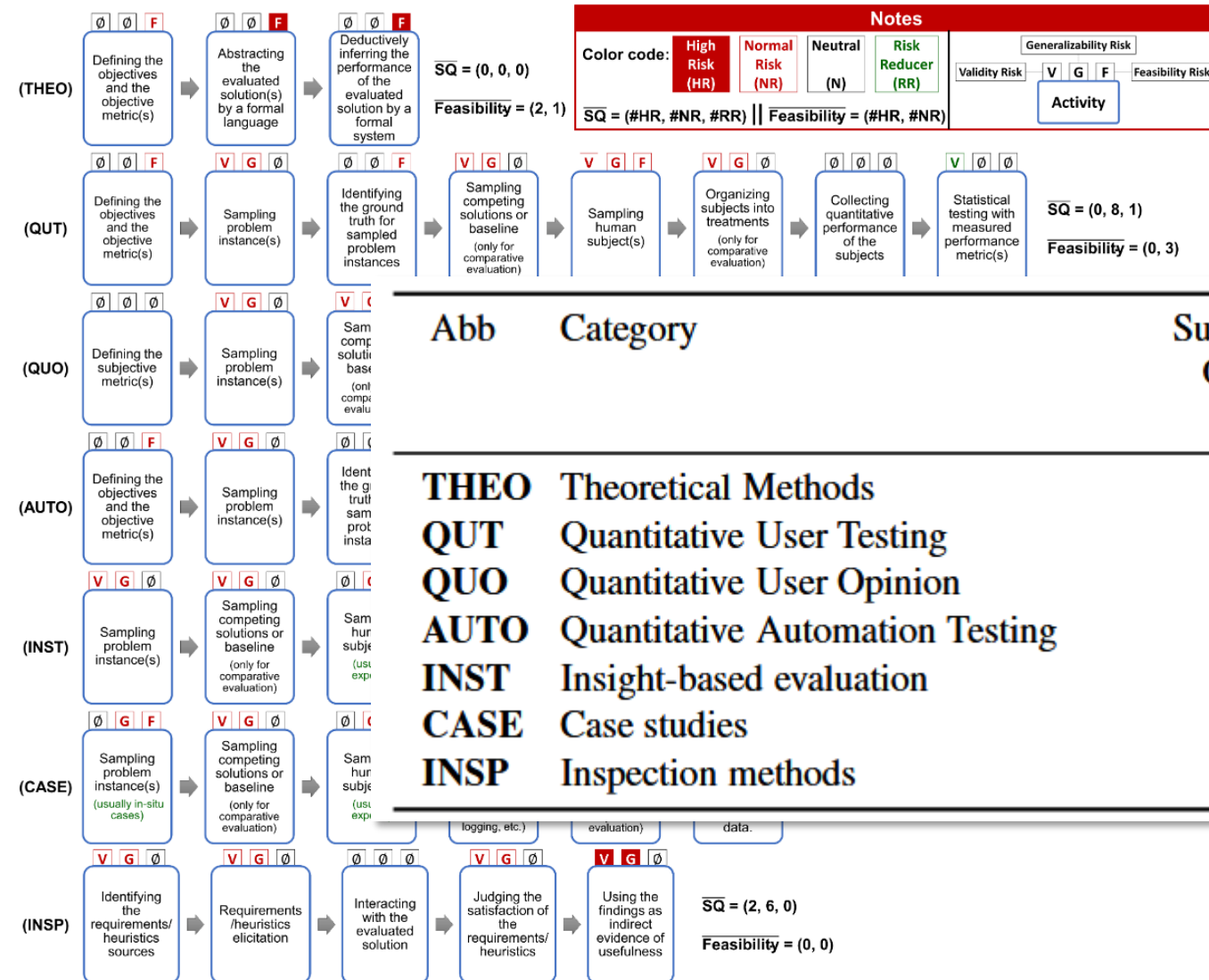
An assembly diagram for each evaluation category:

- Each row is one method.
- Each box is an activity.
- The little markers above a box tell you which criterion is affected by that activity and how risky it is.

Everything summarized into two rollups:

- Summative quality (SQ)
- Feasibility

Summative evaluation methods



An assembly diagram for each evaluation category:

Abb	Category	Summative Quality Rank	Feasibility Rank
THEO	Theoretical Methods	1	6
QUT	Quantitative User Testing	3	4
QUO	Quantitative User Opinion	5	2
AUTO	Quantitative Automation Testing	2	5
INST	Insight-based evaluation	4	2
CASE	Case studies	4	3
INSP	Inspection methods	4	1

one method.
an activity.
markers above a box tell
criterion is affected by
and how risky it is.
summarized into two
quality (SQ)

Summative evaluation methods

- Recommendations:
 1. Always select a feasible method with the highest summative quality.
 2. Provide reasoning for evaluation method choice.
 3. Encouraging insight-based evaluation: *“insight-based evaluation is one of the few methods that do not suffer from high risk factors”*
 4. Apply multiple evaluation methods to minimize risk.

Critiques of VA systems

Wu et al., 2023

- Collects, classifies, and discusses instances of criticisms aimed towards VA systems.
- 257 instances of criticisms.
 - Tagged into 36 types of criticisms.
- Examples:
 - C1: Lacking links of coherence and consistency.
 - C2: Unclear relevance to visual analytics.
 - C3: Vague / over-claimed contributions.
 - C11: Unclear definitions and explanations of data.
 - ...

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IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 29, NO. 1, JANUARY 2023

In Defence of Visual Analytics Systems: Replies to Critics

Aoyu Wu, Dazhen Deng, Furui Cheng, Yingcai Wu, Shixia Liu, and Huamin Qu

Abstract—The last decade has witnessed many visual analytics (VA) systems that make successful applications to wide-ranging domains like urban analytics and explainable AI. However, their research rigor and contributions have been extensively challenged within the visualization community. We come in defence of VA systems by contributing two interview studies for gathering critics and responses to those criticisms. First, we interview 24 researchers to collect criticisms the review comments on their VA work. Through an iterative coding and refinement process, the interview feedback is summarized into a list of 36 common criticisms. Second, we interview 17 researchers to validate our list and collect their responses, thereby discussing implications for defending and improving the scientific values and rigor of VA systems. We highlight that the presented knowledge is deep, extensive, but also imperfect, provocative, and controversial, and thus recommend reading with an inclusive and critical eye. We hope our work can provide thoughts and foundations for conducting VA research and spark discussions to promote the research field forward more rigorously and vibrantly.

Index Terms—Visual Analytics, Theory, Qualitative Study, Design Study, Application, Theoretical and Empirical Research

1 INTRODUCTION

The last decades have witnessed extensive and ever-growing research interests on visual analytics. By combining automated data analysis techniques with interactive visualizations, visual analytics facilitates an effective understanding, reasoning, and decision-making on large and complex datasets [24]. Since the IEEE Conference on Visual Analytics Science and Technology (VAST) was founded in 2006, researchers have contributed many systems for solving complex problems in wide-ranging applications, to which we refer as VA systems. Many systems are outcomes of problem-driven research, whose values have been demonstrated through successful applications in high-impact domains, such as social media [72], sports [2, 73], urban analytics [16, 35], and explainable AI [37, 49]. Research on VA systems have also become an important and impactful research field in visualization (Fig. 1).

Despite the success of those application-oriented VA systems, their research contributions and rigor have been extensively challenged, discussed, and debated [7, 41, 55, 70]. Underlying those debates is the tension between the impetus to create specific software artifacts and the drive of academic research to produce general knowledge [47]. This tension raises the frequently asked question of “what our visualization community can learn from the VA system beyond solving the domain-specific problem.” Furthermore, the design and validation methodology of VA systems is human-centered and thus qualitative and subjective in nature. This nature might contrast with the tropism in science and computer science that embraces quantification and objectivity [23, 41].

The above discrepancies point out some practical problems in the research field. For contributors, their process of planning, developing, validating, and reporting VA systems is prone to diverse mistakes or pitfalls at different stages [55]. The process is challenging because conducting research on VA systems requires a wide range of skills, such as leveraging HCI approaches to understand target users, implementing automated algorithms, and designing effective visual designs [62]. For reviewers, assessing the quality of VA system research requires judging and weighing the above aspects. This assessing process tends to be subjective due to the lack of a shared ground among reviewers regarding

objective criteria for evaluating VA systems [70].

We, therefore, seek to identify a common space of critics for VA systems and corresponding replies to critics. We note that existing instructions on assessing the broad scope of visualization research are not specific to VA systems [33]. Besides, the current reflections and discussions on the vigor of visualization design studies (e.g., [40, 41, 55]) are mainly constructed from authors’ engagement and experiences. While those discussions are undoubtedly critical and insightful, we argue that surveying a broader scope of VA researchers will lead to a more diverse, representative, and convincing understanding.

We conduct two interview studies to reflect on VA systems in terms of their common criticisms and corresponding replies. As shown in Fig. 2, we start by interviewing 24 researchers in charge of 47 VA systems by asking them, “what are the criticisms you have received during peer-reviewing?” We obtain 257 instances of criticisms, which are further classified into 36 common types through iterative categorization. As those criticisms are diverse, deep, and challenging, we conduct another interview study with 17 researchers to validate our classification and gather their replies, e.g., how to mitigate and respond to those criticisms? Drawing upon replies to low-level criticisms, we discuss implications for a high-level question - how to conduct research to defend and improve the research values and rigor of VA systems?

We position our work as a preliminary probe into assessment criteria for VA systems. We expect our work to benefit both VA contributors and reviewers by offering an evidence-based reference for assessing VA systems. We, however, emphasize that criticisms are intrinsically biased, changing, controversial, and fallible, so is our result. It is not our intention to advocate a golden template, but to construct a preliminary and debatable set of criteria for assessing VA systems. We hope our work can provide foundations and spark discussions to make the research field more rigorous and vibrant. We make our interview data and other supplemental material available at [re-vast.github.io](https://github.com/re-vast).

2 RELATED WORK

We focus on research that primarily contributes a VA system for an application. Thus, we review existing literature regarding theoretical advances in visualization application and design study, visual analytics, and empirical methods for understanding the field of visualization.

2.1 Visualization Application and Design Study

Sedlmair et al. [55] formally introduced a visualization design study as “a project in which visualization researchers analyze a specific real-world problem faced by domain experts, design a visualization system that supports solving this problem, validate the design, and reflect about lessons learned in order to refine visualization design guidelines”. Following this definition, they further proposed a nine-stage methodology framework for visualization design studies. Their well-cited framework has become a common method of developing

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Critiques of VA systems

Lacking links of
coherence and
consistency

Introduction

Related work

Problem
abstraction

Data and data
analysis

Visual design

System and
implementation

Evaluation

Others

Critiques of VA systems

C26: Evaluations are incomplete and insufficient

"Therefore, it needs to choose appropriate combinations of evaluation methods to validate the argued contributions."

C27: Unclear evaluation methods and protocols

"... the authors should clarify the participants' demographics, user study procedures, data collection, and coding scheme"

C28: Lacking realism (e.g., real users and datasets)

"However, it is considered far more solid and convincing to evaluate with real domain experts"

C29: Lacking comparison with baseline approaches

"... it is worthy to develop, debate, test, and validate comparative evaluation methods, for example, to break complex VA systems into components and conduct ablation studies (P14), and to compare the final systems with early prototypes (P4)."

C30: Lacking quantitative evaluation and feedback

"Integrating qualitative and quantitative evaluation could make the evaluation more solid and convincing."

C31: Analysis insights are suspicious or not new

"Correspondingly, it degrades the system if the discovered insights are not new or doubtful. The interviewees suggested evaluating with domain experts to gather feedback about the quality of insights and demonstrate how the insights advance the understandings of the domain problem"

C32: Unclear interpretation of expert feedback

"the feedback should be analyzed systematically to provide insightful discussions, especially how the design can be further improved and potentially inform design in related domains."

Supporting design and evaluation of VA systems

- **IVAS** ontology: Improving Visual Analytics Systems
- An ontology that records **symptoms → causes → remedies → side-effects** and supports **abstract reasoning** over a small set of abstract entities built from stats, algorithms, vis, and interaction.
- Designing a VA workflow is viewed as an optimization of the cost-benefit ratio.

Chen and Ebert, 2019

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M. Gleicher, H. Leitte, and I. Viola
(Guest Editors)

Volume 38 (2019), Number 3

An Ontological Framework for Supporting the Design and Evaluation of Visual Analytics Systems

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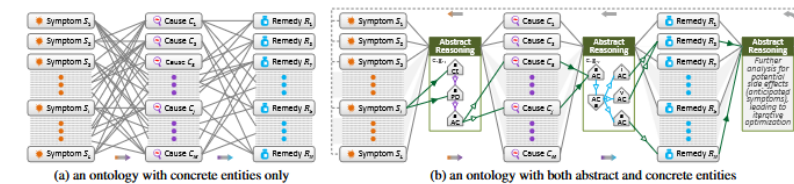


Figure 1: An ontology can support the design and evaluation of visual analytics systems by enabling systematic recording of symptoms, causes, remedies, and side-effects, facilitating the analysis of causal relations, search for solutions, and comparison of different design options. (a) An ontology that records only concrete entities (e.g., instances in practice) will take years and decades to build. (b) An ontology built around abstract entities can enable abstract reasoning in a scalable manner while allowing gradual addition of concrete entities.

Abstract

Designing, evaluating, and improving visual analytics (VA) systems is a primary area of activities in our discipline. In this paper, we present an ontological framework for recording and categorizing technical shortcomings to be addressed in a VA workflow, reasoning about the causes of such problems, identifying technical solutions, and anticipating secondary effects of the solutions. The methodology is built on the theoretical premise that designing a VA workflow is an optimization of the cost-benefit ratio of the processes in the workflow. It makes use of three fundamental measures to group and connect "symptoms", "causes", "remedies", and "side-effects", and guide the search for potential solutions to the problems. In terms of requirement analysis and system design, the proposed methodology can enable system designers to explore the decision space in a structured manner. In terms of evaluation, the proposed methodology is time-efficient and complementary to various forms of empirical studies, such as user surveys, controlled experiments, observational studies, focus group discussions, and so on. In general, it reduces the amount of trial-and-error in the lifecycle of VA system development.

1. Introduction

Designing, evaluating, and improving visual analytics (VA) systems is a primary area of activities in our discipline. Since the first IEEE VAST in 2006, a large number of system papers and application papers have been published, signifying the important contributions to the technology advancements as well as a variety of successful applications [WHC⁺17]. During this relatively short period, VA researchers have made noticeable progress in adopting some best practices in software engineering (e.g., [Sum15]). Today, most system papers and application papers in VA conferences reported user-centered requirement analysis and evaluation. Some

reported the use of incremental, iterative, nested, agile, and evolutionary development approaches [Lar03, Mun09, Mar14].

While these approaches have no doubt successfully delivered many VA systems and applications, assessing and sharing the experience of developing these VA systems and applications can be difficult. These difficulties include:

- The step made from requirements analysis to design specification is typically based on intuitions and the design decisions made are often undocumented, difficult to share, or not reproducible in another application.
- The evaluation based on individual use cases is sometimes per-

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Supporting design and evaluation of VA systems

- **Symptom:** records that describe observable problems in a VA workflow.

One problem here is that S_x

- **Cause:** records causal relations.

S_x may cause S_y

- **Remedy:** Records that describe actions to be taken.

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An Ontological Framework for Supporting the Design and Evaluation of Visual Analytics Systems

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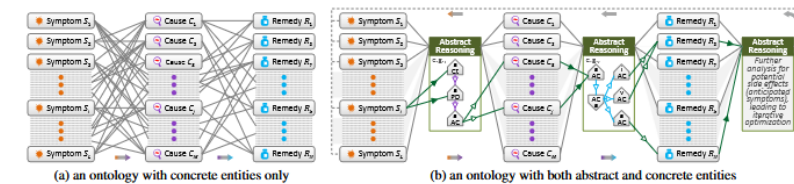


Figure 1: An ontology can support the design and evaluation of visual analytics systems by enabling systematic recording of symptoms, causes, remedies, and side-effects, facilitating the analysis of causal relations, search for solutions, and comparison of different design options. (a) An ontology that records only concrete entities (e.g., instances in practice) will take years and decades to build. (b) An ontology built around abstract entities can enable abstract reasoning in a scalable manner while allowing gradual addition of concrete entities.

Abstract

Designing, evaluating, and improving visual analytics (VA) systems is a primary area of activities in our discipline. In this paper, we present an ontological framework for recording and categorizing technical shortcomings to be addressed in a VA workflow, reasoning about the causes of such problems, identifying technical solutions, and anticipating secondary effects of the solutions. The methodology is built on the theoretical premise that designing a VA workflow is an optimization of the cost-benefit ratio of the processes in the workflow. It makes use of three fundamental measures to group and connect "symptoms", "causes", "remedies", and "side-effects", and guide the search for potential solutions to the problems. In terms of requirement analysis and system design, the proposed methodology can enable system designers to explore the decision space in a structured manner. In terms of evaluation, the proposed methodology is time-efficient and complementary to various forms of empirical studies, such as user surveys, controlled experiments, observational studies, focus group discussions, and so on. In general, it reduces the amount of trial-and-error in the lifecycle of VA system development.

1. Introduction

Designing, evaluating, and improving visual analytics (VA) systems is a primary area of activities in our discipline. Since the first IEEE VAST in 2006, a large number of system papers and application papers have been published, signifying the important contributions to the technology advancements as well as a variety of successful applications [WHC⁺17]. During this relatively short period, VA researchers have made noticeable progress in adopting some best practices in software engineering (e.g., [Sum15]). Today, most system papers and application papers in VA conferences reported user-centered requirement analysis and evaluation. Some

reported the use of incremental, iterative, nested, agile, and evolutionary development approaches [Lar03, Mun09, Mar14].

While these approaches have no doubt successfully delivered many VA systems and applications, assessing and sharing the experience of developing these VA systems and applications can be difficult. These difficulties include:

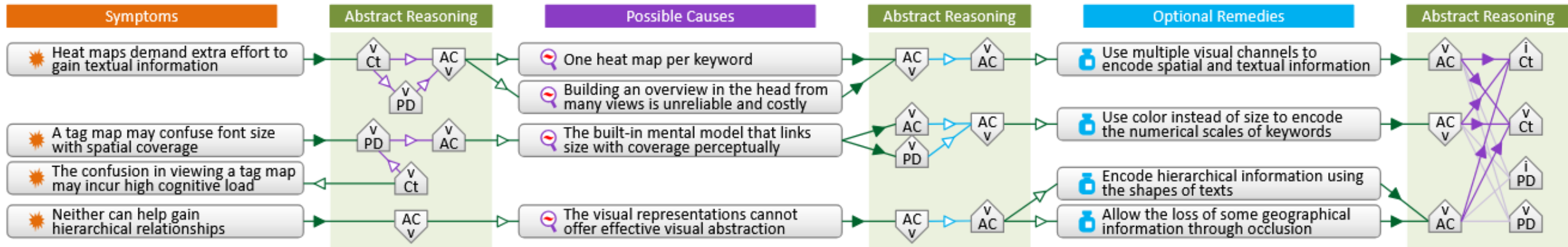
- The step made from requirements analysis to design specification is typically based on intuitions and the design decisions made are often undocumented, difficult to share, or not reproducible in another application.
- The evaluation based on individual use cases is sometimes per-

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- Evaluates how each process transforms its input data space to its output data space in terms of the trade-off among three measures:
 - Alphabet Compression (AC): amount of entropy reduction achieved by a process.
 - Potential distortion (PD): errors typically due to information loss..
 - Cost (C)
- Symptoms, causes, remedies are mapped to these abstractions.

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(c) The abstract reasoning about two empirical studies for evaluating three visual designs for geospatial text visualization [ZSE*18]

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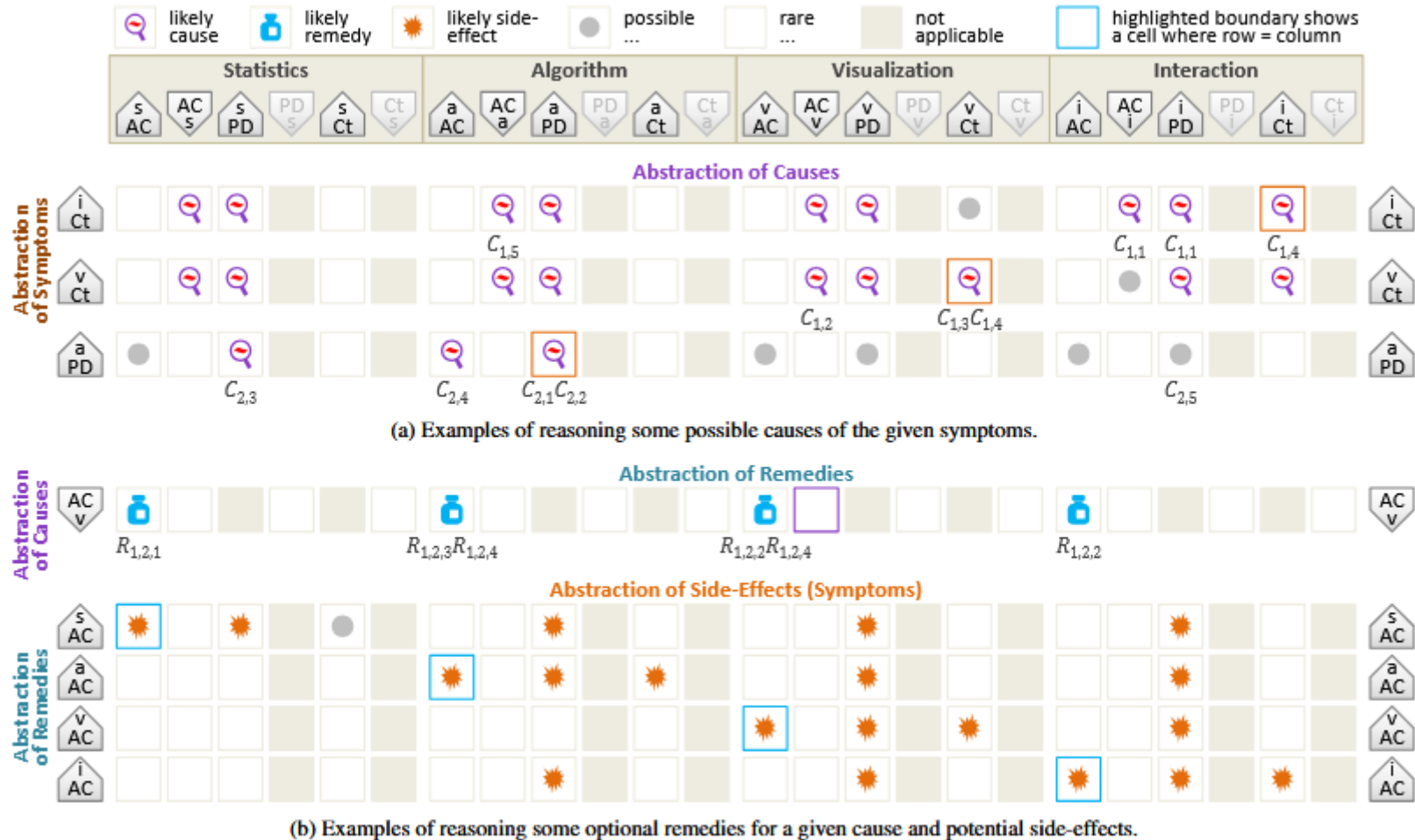


Figure 5: Using the abstract entities, one can conduct abstract reasoning from symptoms to causes, to remedies, and to side-effects in a scalable manner. The rows in this figure are extracted from three 24×24 adjacency matrices in Appendix B.1.

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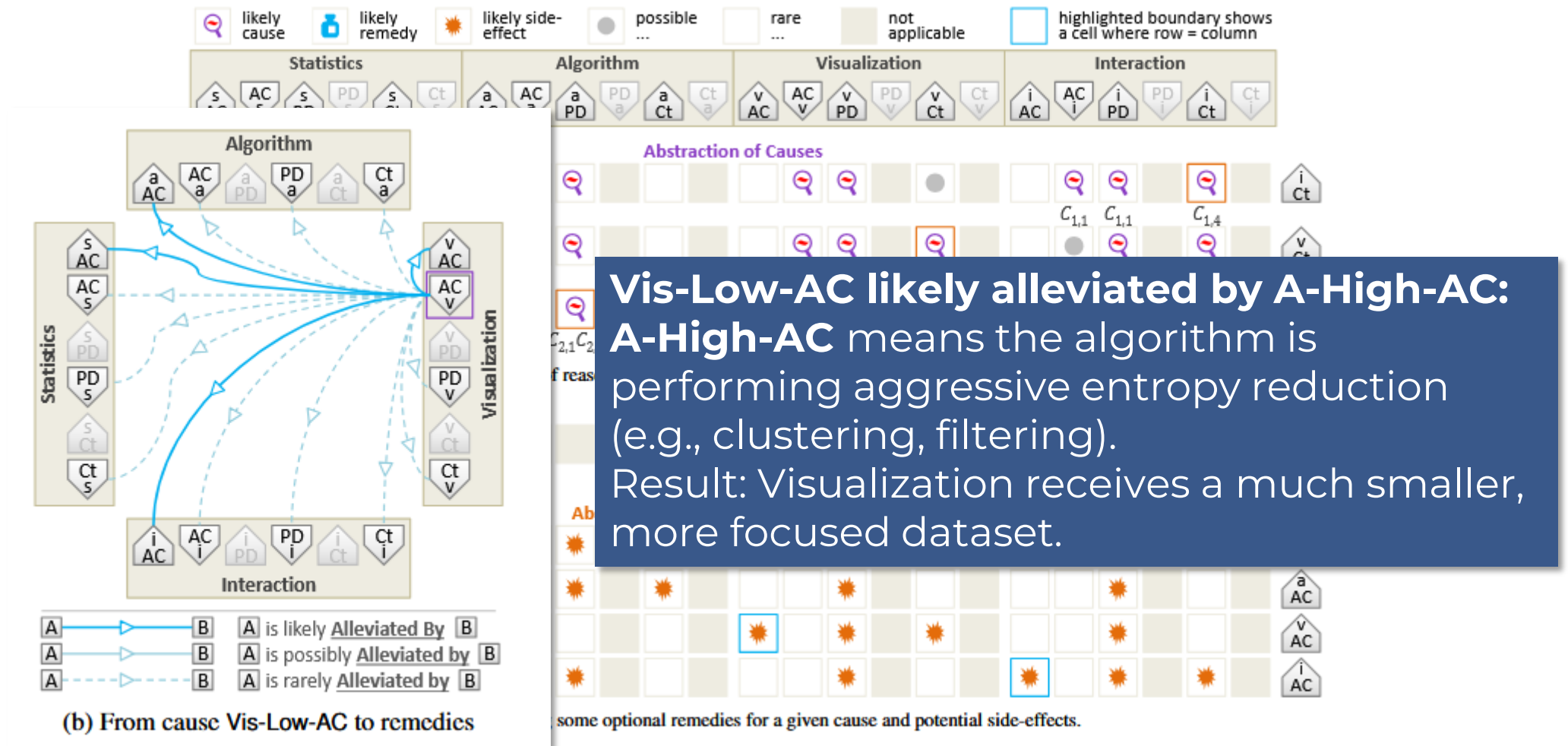


Figure 5: Using the abstract entities, one can conduct abstract reasoning from symptoms to causes, to remedies, and to side-effects in a scalable manner. The rows in this figure are extracted from three 24×24 adjacency matrices in Appendix B.1.

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- Concrete Symptoms → Abstract Reasoning
 - **Action:** You start with a real-world problem, like "the map is too cluttered".
 - **Abstraction:** You map this to one of the 24 abstract entities.
 - *Example:* A cluttered map is often **Vis-Low-AC** (ACv), meaning the visualization isn't compressing/filtering information enough.
- Abstracted Symptoms → Possible Causes
 - **Action:** Using the framework's adjacency matrices, you look for what usually causes that abstract state.
 - **Realization:** You identify the technical reason behind the symptom
- Possible Causes → Abstract Reasoning
 - **Action:** You take that cause and move back into the abstract space to see how to "flip" or alleviate it.
- Abstracted Causes → Optional Remedy
 - **Action:** You look at the suggested abstract remedies and find a concrete technical tool that matches.
- Optional Remedy → Abstract Reasoning (Side-Effects)
 - **Action:** You analyze the chosen remedy to see what *new* problems it might create



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TopoText:

- Technique designed for the visual analytics of multi-scale textual information associated with spatial data.
- Two user studies to evaluate the efficacy of different design alternatives.

TopoText: Context-Preserving Text Data Exploration Across Multiple Spatial Scales

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Figure 1. TopoText showing the prominent topics (encoded by color) at different spatial scales on social media around the city of Keene in the state of New Hampshire, during the Pumpkin Festival riots in 2014. TopoText creates novel text-based visualizations to couple the multi-level textual information in the same visual display for context preservation. (a): The multi-scale boundary-dominant visualization; (b): The multi-scale boundary-space hybrid visualization; (c): The multi-scale space-dominant visualization.

ABSTRACT

TopoText is a context-preserving technique for visualizing text data for multi-scale spatial aggregates to gain insight into spatial phenomena. Conventional exploration requires users to navigate across multiple scales but only presents the information related to the current scale. This limitation potentially adds more steps of interaction and cognitive overload to the users. TopoText renders multi-scale aggregates into a single visual display combining novel text-based encoding and layout methods that draw labels along the boundary or filled within the aggregates. The text itself not only summarizes the semantics at each individual scale, but also indicates the spatial coverage of the aggregates and their underlying hierarchical relationships. We validate TopoText with both a user study as well as several application examples.

ACM Classification Keywords

H.5.m. Information Interfaces and Presentation (e.g. HCI): Miscellaneous

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Author Keywords

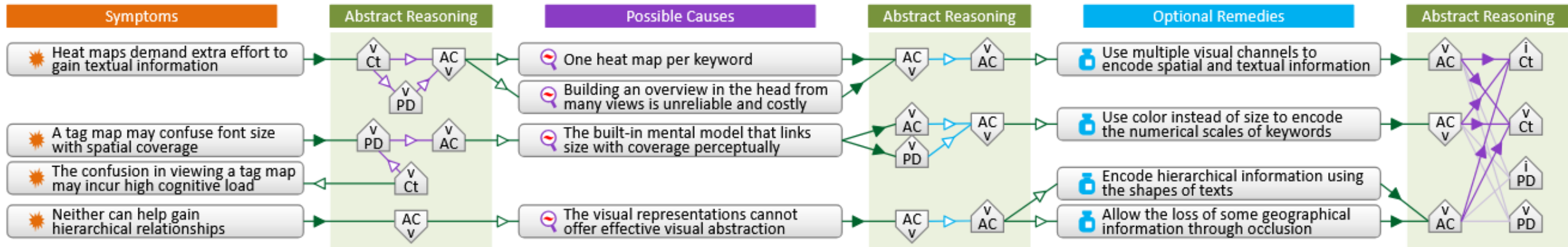
Geospatial visualization; text visualization; typographic map; context preservation; multi-scale analysis.

INTRODUCTION

Spatial data aggregated at different scales often yield different distribution patterns. For example, the dominant criminal offense types in an entire city may be theft and robbery. However, local regions such as a university campus may be dominated by different crimes, i.e., liquor law violation and noise. Thus, there are multiple interpretations of the data depending on scale [36, 52], and acquiring an accurate picture of the city requires understanding them all. Typical approaches to support multi-scale navigation either juxtapose multiple views of each scale for easy comparison [19, 24, 31], combine the analysis results at different scales into a holistic display [26, 51, 57], or focus on optimization to improve navigation efficiency [30, 38, 56]. However, all these approaches either require additional, often significant, interaction to navigate through multiple spatial scales, or yield high visual complexity and clutter. Furthermore, while much spatial data is now textual in nature, such as geotagged social media, most existing techniques are designed for scalar or tensor data.

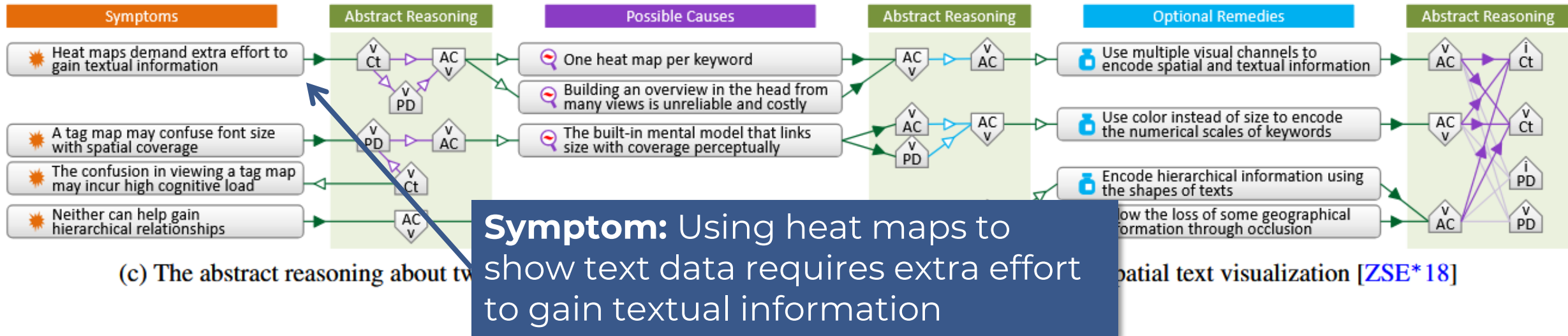
In this paper, we propose TopoText, a technique designed for the visual analysis of multi-scale textual information associated with spatial data, such as geo-tagged social media, crime incident reports, and demographic data. Based on a recent context-preserving visualization of multi-scale spatial aggregates [57], TopoText maximizes screen space utilization and

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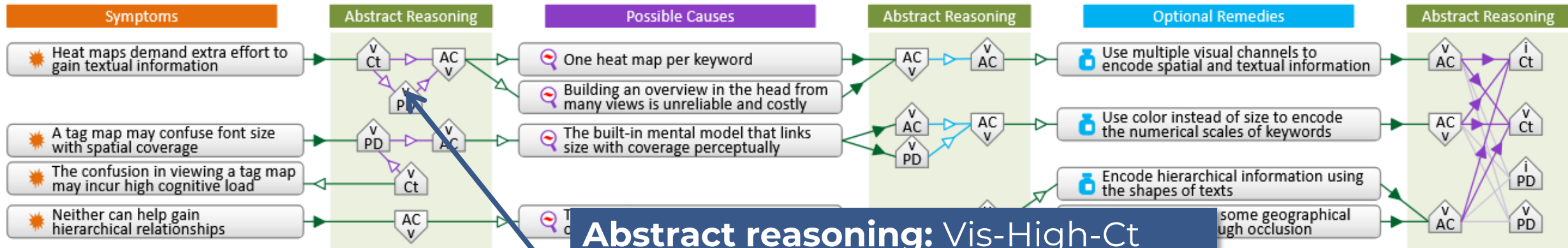


(c) The abstract reasoning about two empirical studies for evaluating three visual designs for geospatial text visualization [ZSE*18]

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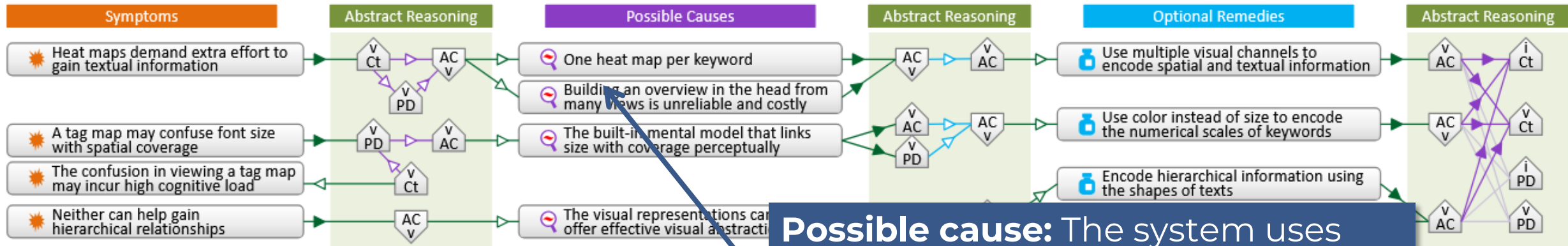


(c) The abstract reasoning about two empiric

Abstract reasoning: Vis-High-Ct meaning there is a high cognitive or time cost for the user to process the visual.

visualization [ZSE*18]

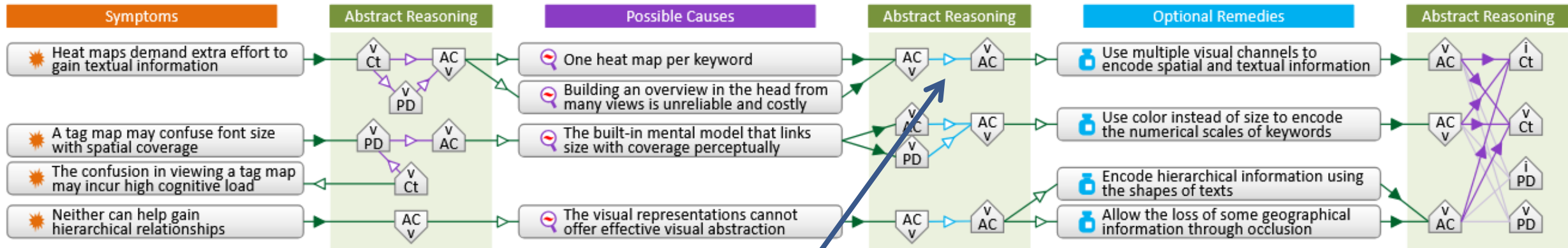
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(c) The abstract reasoning about two empirical studies for evaluation

Possible cause: The system uses "one heat map per keyword," forcing the user to build an overview in their head.

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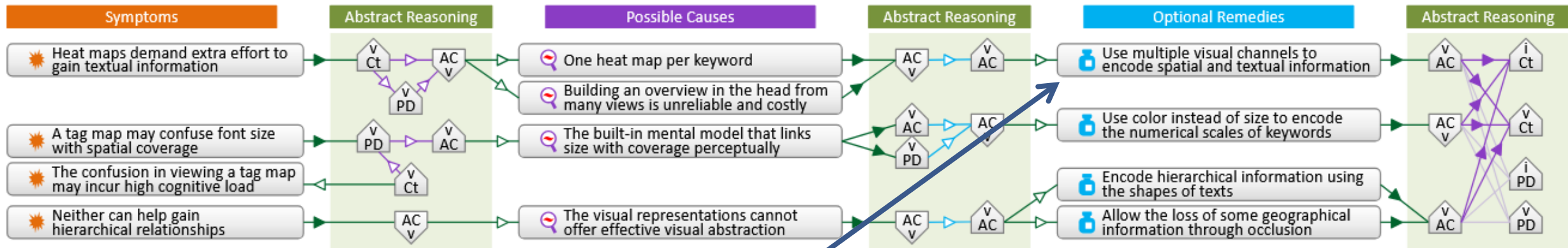


(c) The abstract reasoning about two

Abstract reasoning: Remedy should focus on increasing the amount of AC in the visualization.

ial text visualization [ZSE*18]

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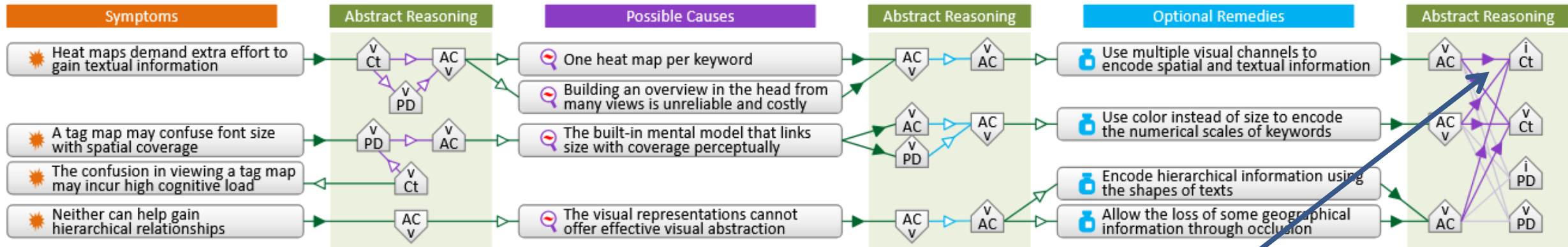


(c) The abstract reasoning about two

Optional remedy: Use "multiple visual channels" to encode both spatial and textual information simultaneously.

ial text visualization [ZSE*18]

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(c) The abstract reasoning about two empirical studies for evaluating three visual designs for geospatial text visualization [ZSE*18]

Abstract reasoning: Remedy may lead to new symptoms, such as increase in interaction cost.